

CPIT-285 | Final Exam Fall 24

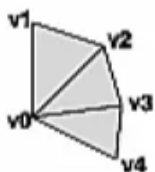
Part 1 Definitions [8 Marks]:

Define the following terms based on what we learned during the semester:

1. Open GL
 2. Rendering
 3. Model Transformation
 4. UV Coordinates
 5. Light Model
 6. Clipping Coordinates
 7. Normal Map
 8. HP/LP / Baking
-

Part 2

1. Determine the number of geometric primitives in the given model : [2 Marks]

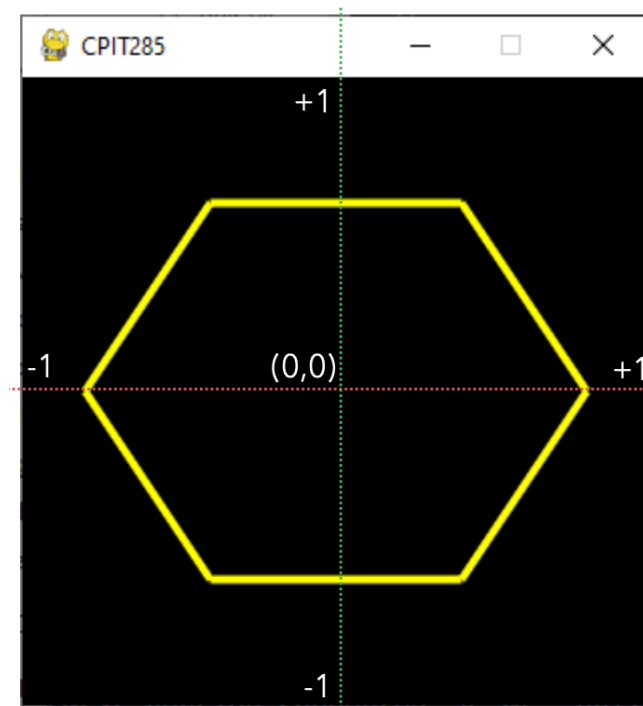


Answer:

2. Using the following OpenGL function, draw the expected output: [3 Marks]

```
gl_Position = vec4(position.x, position.y, position.z, 1.0);
fragColor = vec4(1.0, 1.0, 0.0, 1.0);
GL.glLineWidth(4)
position_data = [[ 0.8,0.0,0.0],
                  [ 0.4,0.6,0.0],
                  [-0.4, 0.6,0.0],
                  [-0.8,0.0,0.0],
                  [-0.4,-0.6,0.0],
                  [ 0.4,-0.6,0.0 ]]
position_attribute = Attribute( 'vec3', position_data)
GL glDrawArrays(GL.GL_LINE_LOOP, 0,
```

Answer:



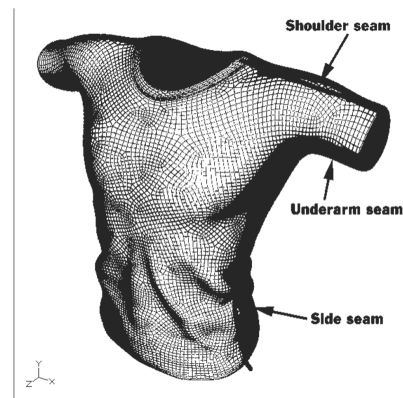
Part 3

1. What is topology, why is it important, and why should I care about it? At what stages can issues caused by poor topology design become apparent? [2 Marks]

Answer:

2. Using the concept of UV unwrapping, unwrap the given T-shirt model. [2 Marks]

Answer:



(صورته مشابهه للي جات)

Q3 According to Abdullah's scene of winning a battle, describe At least five graphical concepts that can be utilized to Enhance the visual quality of the scene.? (لآب الانميشن لوآحد من الشآبآب)

[3 Marks]

Answer:

Part 4: Multiple-Choice Questions [10 Marks]

Q1. Diffuse lighting type is:

a. A light that is scattered by the environment

b. A directional light cast made by a light source ← (تعريف السلايد بس يعتبر غلط)

c. A reflection of light on surfaces

d. None of the above

Q2. What is visual texture?

a. It is the painting of glass objects made to look transparent and shiny

b. The creation of realistic-looking textures in a work

c. The use of delicate brush strokes or other marks to create fine lines . All of the above

d. All the Above

Q3. In which 3D projection do objects that are far away from the camera appear smaller compared to those nearby?

a. Perspective projection

b. Orthographic projection

c. Projection that has no convergence point

d. None of the above

Q4. Ambient lighting type is:

a. A light that is scattered by the environment

b. A directional light cast made by a light source . A reflection of light on surfaces

d. None of the above

Q5. In OpenGL, the clipping area is set by default:

a. -0 To +0

b. -1 To +1

c. -3 To +3

Q6: Which OpenGL primitive is commonly used to draw the faces of a solid 3D cube?

a. GL_LINE_LOOP

b. GL_QUADS

- c. GL_POINTS
- d. GL_TRIANGLES

Q7: Which of the following factors influence a light model in computer graphics?(Question close to this)

- A) Light source properties, such as intensity, color, position, and direction.
- B) Surface attributes, including material properties like color and reflectivity.
- C) Object orientation and the viewer's direction.
- D) All of the above.**

More Additional MCQs. till 10 ...

by : Z

بالتوفيق للجميع، ولا تنسونا من دعواتكم الطيبة .